

Toric Varieties

Labix

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Abstract

Contents

1	Monoids	3
1.1	Basic Properties of Monoids	3
1.2	The Grothendieck Completion of a Group	3
1.3	The Monoid Ring	4
1.4	Affine Semigroup Associated to a Rational Polyhedral Cone	5
2	Introduction to Toric Varieties	6
2.1	The Standard n -Torus	6
2.2	General Toric Varieties	7
3	Constructing Affine Toric Varieties	8
3.1	Equivalent Constructions of Affine Toric Varieties	8
3.2	Cones and Affine Toric Varieties	9
3.3	Properties of Affine Toric Varieties	11
3.4	Morphisms between Affine Toric Varieties	12
4	Constructing Generic Toric Varieties	13
4.1	Fans and Toric Varieties	13
4.2	Orbit Cone Correspondence	14
4.3	Divisors on Toric Varieties	14

1 Monoids

1.1 Basic Properties of Monoids

Definition 1.1.1 (Monoids)

A monoid is a pair $(M, *)$ where M is a set and $*$: $M \times M \rightarrow M$ is a binary operation on M satisfying the following axioms:

- Associativity: $(a * b) * c = a * (b * c)$ for all $a, b, c \in M$.
- Identity: There exists $1_M \in M$ such that $1_M * a = a = a * 1_M$ for all $a \in M$.

Definition 1.1.2 (Monoid Homomorphisms)

Let M, N be monoids. Let $\phi : M \rightarrow N$ be a function. We say that ϕ is a monoid homomorphism if the following are true.

- $\phi(m_1 m_2) = \phi(m_1) \phi(m_2)$ for all $m_1, m_2 \in M$.
- $\phi(1_M) = 1_N$.

Definition 1.1.3 (Commutative Monoids)

Let M be a monoid. We say that M is commutative if

$$a * b = b * a$$

for all $a, b \in M$.

Definition 1.1.4 (Cancellative Monoids)

Let M be a monoid. We say that M is cancellative if the following are true.

- $ab = ac$ implies $b = c$ for all $a, b, c \in M$.
- $ba = ca$ implies $b = c$ for all $a, b, c \in M$.

Definition 1.1.5 (Finitely Generated Monoids)

Let M be a monoid. We say that M is finitely generated if there exists $m_1, \dots, m_n \in M$ such that

$$m = m_1^{a_1} \cdots m_n^{a_n}$$

for all $m \in M$ and some $a_1, \dots, a_n \in \mathbb{N}$.

1.2 The Grothendieck Completion of a Group

Definition 1.2.1 (Grothendieck Completion)

Let M be a commutative monoid. Define the Grothendieck completion of M to be the set

$$G(M) = \frac{M \times M}{\sim}$$

where $(m_1, m_2) \sim (n_1, n_2)$ if $m_1 + n_2 + k = m_2 + n_1 + k$ together with the binary operation $*$: $G(M) \times G(M) \rightarrow G(M)$ defined by

$$(m_1, m_2) * (n_1, n_2) = (m_1 + n_1, m_2 + n_2)$$

Lemma 1.2.2

Let M be a commutative monoid. Then $G(M)$ is an abelian group.

Definition 1.2.3 (The Canonical Homomorphism)

Let M be a commutative monoid. Define the canonical homomorphism to the Grothendieck completion to be the function $i : M \rightarrow G(M)$ given by $i(m) = (m, 0)$.

Proposition 1.2.4 (Universal Property of the Grothendieck Completion)

Let M be a commutative monoid. Then $G(M)$ and the canonical map $i : M \rightarrow G(M)$ satisfies the following universal property: For any abelian group G and any monoid homomorphism $\phi : M \rightarrow G$, there exists a unique group homomorphism $\psi : G(M) \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{i} & G(M) \\ & \searrow \phi & \downarrow \exists! \psi \\ & & G \end{array}$$

Moreover, $G(M)$ is the unique (up to unique isomorphism) abelian group satisfying this property.

Lemma 1.2.5

Let M be a cancellative commutative monoid. Then the canonical morphism $i : M \rightarrow G(M)$ is injective.

1.3 The Monoid Ring

Definition 1.3.1 (The Monoid Ring)

Let S be a monoid. Let R be a ring. Define the semi-group ring of S over R to be the left R -module

$$R[S] = \bigoplus_{s \in S} R \cdot t^s$$

together with multiplication defined on basis by $t^{s_1} \cdot t^{s_2} = t^{s_1+s_2}$

Lemma 1.3.2

Let S be a monoid. Let R be a ring. Then the following are true.

- $R[S]$ is a ring.
- If S is commutative, then $R[S]$ is a commutative ring.
- If R and S are commutative, then $R[S]$ is an R -algebra.

Definition 1.3.3 (Monomial Notation)

Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. Define

$$x^u = x_1^{u_1} \cdots x_n^{u_n} \in R[x_1, \dots, x_n]$$

Definition 1.3.4 (Lattice Ideal)

Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Define the lattice ideal of L in $R[x_1, \dots, x_n]$ to be

$$I_L = \langle x^u - x^v \mid u, v \in \mathbb{N}^n \text{ and } u - v \in L \rangle$$

Proposition 1.3.5

Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S . Let $\phi : \mathbb{Z}^n \rightarrow G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then there is a k -algebra isomorphism

$$k[S] \cong \frac{k[x_1, \dots, x_n]}{I_{\ker(\phi)}}$$

Proposition 1.3.6

Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S . Then we have

$$\dim(k[S]) = n - \text{rank}(\ker(\phi))$$

Definition 1.3.7 (Affine Semi-group)

Let S be a monoid. We say that S is an affine semi-group if there exists an injective monoid homomorphism $S \rightarrow \mathbb{N}^d$ for some $d \in \mathbb{N}$.

Proposition 1.3.8

Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S . Let $\phi : \mathbb{Z}^n \rightarrow G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then the following are equivalent.

- $I_{\ker(\phi)}$ is prime.
- $k[S]$ is an integral domain.
- $G(S)$ is a free abelian group.
- S is an affine semi-group.

1.4 Affine Semigroup Associated to a Rational Polyhedral Cone**Lemma 1.4.1** (Gordon's Lemma)

Let $\sigma \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Let $A \subseteq \mathbb{Z}^n$ be a subgroup. Then $\sigma \cap A$ is an affine semi-group.

Conversely, given an affine semigroup $S \subseteq \mathbb{N}^n$, we may associate to it a cone

$$\mathbb{R}_{\geq 0}S = \{rs \mid r \in \mathbb{R}_{\geq 0} \text{ and } s \in S\}$$

So given an affine semigroup, we may obtain a new affine semigroup by passing to cones then passing back. This is called the saturation of the affine semigroup.

Definition 1.4.2 (Saturation of an Affine Semigroup)

Let S be an affine semigroup. Define the saturation of S to be

$$S_{\text{sat}} = (\mathbb{R}_{\geq 0}S) \cap G(S)$$

Proposition 1.4.3

Let S be an affine semigroup. Then we have

$$\overline{k[S]} = k[S_{\text{sat}}]$$

Definition 1.4.4 (Saturated Affine Semigroup)

Let S be an affine semigroup. We say that S is saturated if $S = S_{\text{sat}}$.

Lemma 1.4.5

Let $\sigma \subseteq \mathbb{R}^n$ be a strongly convex rational polyhedral cone. Then $\sigma \cap S$ is saturated.

Lemma 1.4.6

Let $\sigma_1, \sigma_2 \subseteq \mathbb{R}^n$ be cones such that $\sigma_1 \cap \sigma_2$ is a face of both cones. Let $A \subseteq \mathbb{Z}^n$ be a subgroup. Then we have

$$(\sigma_1 \cap \sigma_2) \cap A = (\sigma_1 \cap A) + (\sigma_2 \cap A)$$

2 Introduction to Toric Varieties

2.1 The Standard n -Torus

Definition 2.1.1 (The Standard n -Torus)

Let k be a field. Let $n \in \mathbb{N}^+$. Define the standard n -torus over k to be the affine group variety

$$\mathbb{G}_m^n(k) = \text{Spec}(k[x, x^{-1}])^n$$

Definition 2.1.2 (Tori)

Let k be a field. Let T be an affine variety over k . We say that T is a torus if there exists a field extension K/k such that $T \cong \mathbb{G}_m^n(K)$.

Definition 2.1.3 (Split Tori)

Let k be a field. Let T be a torus over the field extension K/k . We say that T is split if $K = k$.

If K is separably closed or algebraically closed, then all tori are split.

Lemma 2.1.4

Let k be a field. Let T be a torus over k . Then the following are true.

- Let T' be a torus over k . Let $\phi : T \rightarrow T'$ be a morphism of group schemes. Then $\text{im}(\phi)$ is closed torus in T' .
- Let $H \leq T$ be a subgroup variety of T . If H is irreducible, then H is a torus.
- Let $H \leq T$ be a subgroup of T . Then T/H is a torus.

Recall that the functor $D : \mathbf{FGAb} \rightarrow \mathbf{Diag}$ sending a finitely generated abelian group G to $D(G) = \text{Spec}(k[G])$ is an equivalence of categories. Evidently split tori corresponds to the free abelian groups of finite rank under the equivalence.

The following are some more properties of a split torus T :

- T is diagonalizable (T is a torus if and only if T is smooth connected diagonalizable)
- Every representation of G is diagonalizable.
- If V is a representation of T , then we can decompose V into its eigenspaces:

$$V = \bigoplus_{\chi \in \chi(T)} V_\chi$$

Recall that the character of a torus T is a group homomorphism $T \rightarrow \mathbb{C}^\times$.

Definition 2.1.5 (Character Lattices)

Let k be a field. Let T be a torus over k . Define the character lattice of T to be the group of characters

$$\chi(T) = \text{Hom}_{\mathbf{GrpSch}_k}(T, \mathbb{G}_m)$$

Let $(m_1, \dots, m_n) \in \mathbb{Z}^n$, we have a character $\chi : T \rightarrow \mathbb{C}^\times$ defined by $(t_1, \dots, t_n) \mapsto t_1^{m_1} \cdots t_n^{m_n}$. All characters of T are of this form. In other words, we have $M = \text{Hom}(T, \mathbb{C}^\times) \cong \mathbb{Z}^n$.

Definition 2.1.6 (One-parameter Subgroups)

Let k be a field. Let T be a torus over k . A one-parameter subgroup of T is a homomorphism of group schemes $\mathbb{G}_m \rightarrow T$.

Since χ is the inverse to D , we have the following result.

Lemma 2.1.7

Let k be a field. Let T be a split torus over k . Then the following are true.

- The map

$$\chi(T) \times \operatorname{Hom}_{\mathbf{GrpSch}_k}(\mathbb{G}_m, T) \rightarrow \mathbb{Z}$$

defined by $(\chi, \lambda) \rightarrow \chi \circ \lambda$ is a perfect pairing.

- A choice of an isomorphism $T \cong \mathbb{G}_m^n$ induces an isomorphism $\chi(T) \cong \mathbb{Z}^n \cong \operatorname{Hom}_{\mathbf{GrpSch}_k}(\mathbb{G}_m, T)$ such that the above perfect pairing is the standard Euclidean dot product.

2.2 General Toric Varieties

Recall that the t functor $t : \mathbf{Var}_k \rightarrow \mathbf{Sch}_k$ is the fully faithful functor whose inverse restricted to the essential image sends a scheme X to its underlying k -points $X(k)$. When $X = \operatorname{Spec}(\mathbb{C}[x_1, \dots, x_n]/I)$ is affine over $\mathbb{C}$ then the $\mathbb{C}$ -points are precisely $\mathbb{V}(I)$. When X is projective over $\mathbb{C}$ then its $\mathbb{C}$ -points are described locally as the $\mathbb{C}$ -points on the affine varieties in each standard affine cover. Thus we freely interchange notations between scheme theory and classical algebraic geometry.

Definition 2.2.1 (Toric Varieties)

Let X be an irreducible variety over $\mathbb{C}$. We say that X is a toric variety if the following are true.

- X contains a torus T_X as a Zariski open subset.
- The action of T_X on itself extends to X .

Example 2.2.2

The following are true.

- $\mathbb{C}^\times$ is a toric variety.
- $\mathbb{C}^n$ is a toric variety.
- $\mathbb{P}^n$ is a toric variety.
- If X, Y are toric varieties, then $X \times Y$ is a toric variety.
- The total space of the line bundle over $\mathbb{P}^n$ is a toric variety.

The equivariant structure of the torus encodes a lot of the algebraic geometry into combinatorial data or more manageable structure. A lot of common varieties are toric, or are subvarieties of complete intersections of toric varieties.

There are two ways to construct normal toric varieties:

(1): Make an open cover of it via affine schemes that each corresponds to some polyhedral cone, then glue the open cover from combinatorial data.

(2): Encode a projective variety as the proj construction given by a polytope.

3 Constructing Affine Toric Varieties

3.1 Equivalent Constructions of Affine Toric Varieties

Definition 3.1.1 (Affine Toric Varieties Associated to Characters)

Let T be a torus over $\mathbb{C}$. Let $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$ be characters of T . Define the affine toric variety associated to $\mathcal{A}$ to be

$$Y_{\mathcal{A}} = \overline{\text{im}(\Phi_{\mathcal{A}})}$$

where $\Phi_{\mathcal{A}} : T \rightarrow \mathbb{A}_{\mathbb{C}}^s$ is the map defined by $\Phi_{\mathcal{A}}(t) = (\chi_1(t), \dots, \chi_s(t))$

Proposition 3.1.2

Let T be a torus over $\mathbb{C}$. Let $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$ be characters of T . Then the following are true.

- $Y_{\mathcal{A}}$ is an affine toric variety.
- $\chi(T_{Y_{\mathcal{A}}}) = \langle \mathcal{A} \rangle$.
- $\dim(Y_{\mathcal{A}}) = \text{rank}(\langle \mathcal{A} \rangle)$.
- The ideal of $Y_{\mathcal{A}}$ is given by

$$\mathbb{I}(Y_{\mathcal{A}}) = I_{\ker(\phi)} = \langle x^{\alpha} - x^{\beta} \mid \alpha, \beta \in \mathbb{N}^s, \alpha - \beta \in \ker(\phi) \rangle$$

where $\phi : \mathbb{Z}^s \rightarrow \chi(T)$ is the group homomorphism defined by $e_i \mapsto \chi_i$.

Proof

Notice that the image of $\Phi_{\mathcal{A}}$ lands in $\mathbb{G}_m^s$. Hence $\Phi_{\mathcal{A}}$ is a map of torii. By lmm we know that $\text{im}(\Phi_{\mathcal{A}})$ is a closed torus in $\mathbb{G}_m^s$. By the relative closure formula, we have that

$$Y_{\mathcal{A}} \cap \mathbb{G}_m^s = \text{im}(\Phi_{\mathcal{A}})$$

Then by definition of subspace topology on $Y_{\mathcal{A}}$, we conclude that $\text{im}(\Phi_{\mathcal{A}})$ is an open subset of $Y_{\mathcal{A}}$. This shows that $Y_{\mathcal{A}}$ contains a torus as an open subset. Also, since $\text{im}(\Phi_{\mathcal{A}})$ is irreducible and is an open subset of $Y_{\mathcal{A}}$, we conclude that $Y_{\mathcal{A}}$ is irreducible.

Since T acts on itself for being a torus, the map $\Phi_{\mathcal{A}}$ gives the action of $\text{im}(\Phi_{\mathcal{A}})$ on $\mathbb{G}_m^s$ that sends varieties to varieties. Let $t \in \text{im}(\Phi_{\mathcal{A}})$. Since $\text{im}(\Phi_{\mathcal{A}})$ is a torus, we have that

$$\text{im}(\Phi_{\mathcal{A}}) = t \cdot \text{im}(\Phi_{\mathcal{A}}) \subseteq t \cdot Y_{\mathcal{A}}$$

By definition of closure, we must have $Y_{\mathcal{A}} \subseteq t \cdot Y_{\mathcal{A}}$. Replacing t with t^{-1} show that $t \cdot Y_{\mathcal{A}} \subseteq Y_{\mathcal{A}}$. Hence the action of $\text{im}(\Phi_{\mathcal{A}})$ gives an action on $Y_{\mathcal{A}}$. Thus $Y_{\mathcal{A}}$ is an affine toric variety. ■

Example 3.1.3 (The Rational Normal Cone)

Let $d \in \mathbb{N}^+$. Let $\phi : \mathbb{C}^2 \rightarrow \mathbb{C}^{d+1}$ be the morphism defined by

$$\phi(s, t) = (s^d, s^{d-1}t, \dots, st^{d-1}, t^d)$$

Then the following are true.

- $\text{im}(\phi)$ is an affine toric variety.
- $\mathbb{I}(\text{im}(\phi)) = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d)$.

Proof

Let $I = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d) \subseteq \mathbb{C}[x_1, \dots, x_{d+1}]$. Notice that by projecting to projective space, we have that ϕ is the veronese map. It follows from the same technique that $\text{im}(\phi) = \mathbb{V}(I)$. By the above, we know that $\text{im}(\phi)$ is an affine toric variety.

Now we have $\mathbb{I}(\text{im}(\phi)) = \mathbb{I}(\mathbb{V}(I)) = I$ because toric varieties are irreducible. ■

Definition 3.1.4 (Toric Ideals)

Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Let $I_L \subseteq R[x_1, \dots, x_n]$ be the lattice ideal of L . We say that I_L is a toric ideal if I_L is a prime ideal.

Lemma 3.1.5

Let $I \subseteq \mathbb{C}[x_1, \dots, x_n]$ be an ideal. Then I is toric if and only if it is prime and is generated by binomials of the form $x_{i_1}^{\alpha_1} \cdots x_{i_n}^{\alpha_n} - x_{j_1}^{\beta_1} \cdots x_{j_m}^{\beta_m}$.

Proof

If I is a toric ideal then it is clearly prime and generated by binomials. So suppose that I is prime and is generated by binomials of the above form. Notice that $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$ is non-empty since it contains $(1, \dots, 1)$. We check that it is a subgroup of $(\mathbb{C}^\times)^n$. Let $u, v \in \mathbb{V}(I) \cap (\mathbb{C}^\times)^n$. Then we know that $u^\alpha - u^\beta = 0 = v^\alpha - v^\beta$. Then we have

$$(uv)^\alpha - (uv)^\beta = u^\alpha v^\alpha - u^\beta v^\alpha + u^\beta v^\alpha - u^\beta v^\beta = (u^\alpha - u^\beta)v^\alpha + u^\beta(v^\alpha - v^\beta) = 0$$

Hence products are closed. Also, we have Hence inverses are closed. Thus $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$ is a subgroup of $(\mathbb{C}^\times)^n$. By Imm, we conclude that $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$ is a torus. For each $1 \leq i \leq n$, projecting to the i th coordinate gives a character $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n \rightarrow (\mathbb{C}^\times)^n \rightarrow \mathbb{C}^\times$. Let $\mathcal{A}$ be the set of these characters. Then by construction we have $\mathbb{V}(I) = Y_{\mathcal{A}}$. By the Nullstellensatz, we have $I = \mathbb{I}(\mathbb{V}(I)) = \mathbb{I}(Y_{\mathcal{A}})$. Hence by the above we conclude that I is toric. ■

Proposition 3.1.6

Let S be an affine semigroup. Then the following are true.

- $\text{Spec}(\mathbb{C}[S])$ is an affine toric variety.
- The character group of the torus in $\text{Spec}(\mathbb{C}[S])$ is given by $\langle t^s \mid s \in S \rangle$.

Proposition 3.1.7

Let X be an irreducible affine variety over $\mathbb{C}$. Then the following are equivalent.

- X is a affine toric variety.
- There exists a torus T and a finite subset $\mathcal{A} \subseteq \chi(T)$ such that $X = Y_{\mathcal{A}}$.
- There exists a toric ideal I of $\mathbb{C}[x_1, \dots, x_n]$ such that $X \cong \text{Spec}\left(\frac{\mathbb{C}[x_1, \dots, x_n]}{I}\right)$.
- There exists an affine semigroup S such that $X \cong \text{Spec}(\mathbb{C}[S])$.

3.2 Cones and Affine Toric Varieties

By the main result in the last section, there is a close connection between affine toric varieties and affine semigroups. In combinatorial commutative algebra we have seen how to construct an affine semigroup from cones and vice versa. We will investigate the connection between cones and affine toric varieties here.

Definition 3.2.1 (Affine Semigroup Associated to a Cone)

Let $\sigma \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Define the associated affine semigroup to be

$$S_\sigma = \sigma^\vee \cap (\mathbb{R}^n)^*$$

Example 3.2.2

There is an isomorphism

$$\text{Spec}(\mathbb{C}[\{0\}^\vee \cap \mathbb{Z}^n]) \cong \text{Spec}(\mathbb{G}_m^n)$$

Proof

The dual cone of $\{0\}$ is $\mathbb{R}^n$ and intersecting with $\mathbb{Z}^n$ just gives itself. So $\mathbb{C}[\mathbb{Z}^n] = \mathbb{C}[x_1, \dots, x_n, x_1^{-1}, \dots, x_n^{-1}]$ is just the coordinate ring of the standard n -torus. ■

Example 3.2.3

Let $e_1, e_2, e_3 \in \mathbb{R}^3$ be the standard basis vectors. Let $\sigma = \text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3) \subseteq \mathbb{R}^3$. Then we have

$$\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$$

Proof

We compute the dual cone to be

$$\sigma^\vee = H_{(1,0,0)}^+ \cap H_{(0,1,0)}^+ \cap H_{(1,0,1)}^+ \cap H_{(0,1,1)}^+$$

A generating set of S_σ as a monoid is given by $\{e_1, e_2, e_3, e_1 + e_2 - e_3\}$. The kernel of the map $\mathbb{Z}^4 \rightarrow \mathbb{Z}^3$ defined by $e_i \mapsto e_i$ for $1 \leq i \leq 3$ and $e_4 \mapsto (1, 1, -1)$ is given by $\mathbb{Z}\langle e_1 + e_2 - e_3 - e_4 \rangle$. Thus we are done. ■

Example 3.2.4

Let $e_1, e_2, e_3 \in \mathbb{R}^3$ be the standard basis vectors. Let $\sigma = \text{Cone}(e_2, e_3, 2e_1 + e_2 + e_3) \subseteq \mathbb{R}^3$. Then we have

$$\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x_1, \dots, x_6]}{(x_1x_4 - x_2x_3, x_1x_5 - x_2^2, x_1x_6 - x_3^2)}$$

Proof

We compute the dual cone to be

$$\sigma^\vee = H_{(2,1,1)}^+ \cap H_{(0,1,0)}^+ \cap H_{(0,0,1)}^+$$

A generating set of S_σ as a monoid is then given by $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (-1, 1, 1), (-1, 2, 0), (-1, 0, 2)\}$. The kernel of the map $\mathbb{Z}^6 \rightarrow \mathbb{Z}^3$ defined by $e_i \mapsto e_i$ for $1 \leq i \leq 3$ and $e_4 \mapsto (-1, 1, 1), e_5 \mapsto (-1, 2, 0), e_6 \mapsto (-1, 0, 2)$ is given by $\mathbb{Z}\langle e_1 - e_2 - e_3 + e_4, e_1 - 2e_2 + e_5, e_1 - 2e_3 + e_6 \rangle$. The ideal associated to this lattice is given by $(x_1x_4 - x_2x_3, x_1x_5 - x_2^2, x_1x_6 - x_3^2)$. Thus we are done. ■

Example 3.2.5

Let $e_1, e_2 \in \mathbb{R}^2$ be the standard basis vectors. Let $\sigma = \text{Cone}(de_1 - e_2, e_2)$. Then we have

$$\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x_0, \dots, x_d]}{(x_i x_{j+1} - x_j x_{i+1} \mid 0 \leq i < j \leq d-1)}$$

Proof

The dual cone is given by

$$\sigma^\vee = H_{(d,-1)}^+ \cap H_{(0,1)}^+$$

A generating set for the monoid S_σ is given by $\{e_1 + ie_2 \mid 0 \leq i \leq d\}$. The kernel of the map $\mathbb{Z}^{d+1} \rightarrow \mathbb{Z}^2$ defined by $e_i \mapsto e_1 + ie_2$ is given by $\mathbb{Z}\langle e_i + e_{j+1} - e_{i+1}e_j \mid 0 \leq i < j \leq d-1 \rangle$. Notice that the lattice ideal of the kernel is precisely the ideal defining the rational normal cone. Hence we are done. ■

Not all affine toric varieties comes from cones. But all affine normal toric varieties come from strongly convex rational polyhedral cones.

Proposition 3.2.6

Let V be an affine toric variety. Then the following are equivalent.

- V is normal.
- $V = \text{Spec}(\mathbb{C}[S])$ where S is a saturated affine semigroup.
- $V = \text{Spec}(\mathbb{C}[S_\sigma])$ where σ is a strongly convex rational polyhedral cone.

Let S be an affine semigroup. Recall that the saturation of S is the affine semigroup

$$S_{\text{sat}} = \mathbb{R}_{\geq 0}S \cap G(S)$$

where $G(S)$ is the Grothendieck completion of S .

Lemma 3.2.7

Let $X = \text{Spec}(\mathbb{C}[S])$ be an affine toric variety over $\mathbb{C}$ where S is an affine semigroup. Then the normalization of X is given by

$$\tilde{X} = \text{Spec}(\mathbb{C}[S_{\text{sat}}])$$

together with the normalization map given by the inclusion $\mathbb{C}[S] \hookrightarrow \mathbb{C}[S_{\text{sat}}]$.

Proposition 3.2.8

Let V be an affine toric variety. Then V is smooth if and only if $V = \text{Spec}(\mathbb{C}[S_\sigma])$ where σ is a strongly convex regular rational polyhedral cone.

Thus we have constructed a dictionary

$$\begin{array}{ll} \text{Affine Toric Varieties} & \xleftrightarrow{1:1} \text{Rational Polyhedral Cones} \\ \text{Affine Normal Toric Varieties} & \xleftrightarrow{1:1} \text{Strongly Convex Rational Polyhedral Cones} \\ \text{Affine Smooth Toric Varieties} & \xleftrightarrow{1:1} \text{Regular Rational Polyhedral Cones} \\ \text{Affine Toric Orbifolds} & \xleftrightarrow{1:1} \text{Simplicial Rational Polyhedral Cones} \end{array}$$

3.3 Properties of Affine Toric Varieties**Proposition 3.3.1**

Let k be an algebraically closed field. Let S be an affine semi-group with generators given by $m_1, \dots, m_s$. Then there is a bijection

$$\{p \in \text{Spec}(k[S]) \mid p \text{ is a closed point}\} \xleftrightarrow{1:1} \{\phi : S \rightarrow (\mathbb{C}, \cdot) \mid \phi \text{ is a monoid homomorphism}\}$$

given as follows.

- Each closed point $p \in \text{Spec}(k[S])$ corresponds to the monoid homomorphism $\gamma_p : S \rightarrow \mathbb{C}$ defined by

$$\gamma_p \left(\sum_{i=1}^s \lambda_i m_i \right) = \sum_{i=1}^s \lambda_i x_i(p)$$

where $x_i \in \mathbb{C}[S]$ is the i th coordinate function.

- Each monoid homomorphism $\gamma : S \rightarrow (\mathbb{C}, \cdot)$ corresponds to the point $(\gamma(m_1), \dots, \gamma(m_s))$.

Proposition 3.3.2

Let $n \in \mathbb{N}^+$. Let $\sigma \subseteq \mathbb{R}^n$ be a cone. Let $\tau \subseteq \sigma$ be a face of the cone. Let H_ϕ be the supporting hyperplane for τ where $\phi \in \sigma^\vee \cap \mathbb{Z}^n$. Then we have

$$\mathbb{C}[S_\tau] = \mathbb{C}[S_\sigma]_\phi$$

Example 3.3.3

Let $e_1, e_2, e_3 \subseteq \mathbb{R}^3$ be the standard basis vectors. Let $\sigma = \text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3) \subseteq \mathbb{R}^3$. Let $\tau = \text{Cone}(e_1, e_1 + e_3)$. Then we have

$$\mathbb{C}[S_\tau] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - 1)}$$

Proof

We already computed that $\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$. Notice that the supporting hyperplane for τ is given by $H_{(0,1,0)}^+$ (cross product of e_1 and $e_1 + e_3$ gives the normal to the plane $\mathbb{R}\langle e_1, e_1 + e_3 \rangle$). Under the isomorphism $\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$, the element $(0, 1, 0) \in S_\sigma$ corresponds to y . Hence

$$\mathbb{C}[S_\tau] = \left(\frac{\mathbb{C}[x, y, z, w]}{(xy - zw)} \right)_y \cong \mathbb{C}[y, z, w, y^{-1}] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - 1)}$$

On the other hand, we may compute the coordinate ring $\mathbb{C}[S_\tau]$ using the lattice ideal. Notice that the dual of τ is given by

$$\tau^\vee = H_{(1,0,0)}^+ \cap H_{(1,0,1)}^+$$

A generating set for the monoid S_τ is given by $\{e_2, -e_2, e_3, e_1 - e_3\}$. Then the kernel of the map $\mathbb{Z}^4 \rightarrow \mathbb{Z}^3$ defined by $e_i \mapsto e_i$ for $2 \leq i \leq 3$ and $e_4 \mapsto -e_2, e_1 \mapsto e_1 - e_3$ is given by $\mathbb{Z}\langle e_2 + e_4 \rangle$ and so the coordinate ring is given by

$$\mathbb{C}[S_\tau] = \frac{\mathbb{C}[x, y, z, w]}{(yw - 1)}$$

and the two computations give isomorphic $\mathbb{C}$ -algebras. ■

3.4 Morphisms between Affine Toric Varieties

4 Constructing Generic Toric Varieties

4.1 Fans and Toric Varieties

Definition 4.1.1 (Scheme Associated to a Fan)

Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Define the scheme X_Σ associated to Σ by the following gluing data.

- The schemes $\text{Spec}(\mathbb{C}[S_\sigma])$ for each $\sigma \in \Sigma$.
- For each $\sigma_1, \sigma_2 \in \Sigma$, an open subscheme

$$\text{Spec}(\mathbb{C}[S_{\sigma_1 \cap \sigma_2}])$$

of $\text{Spec}(\mathbb{C}[S_{\sigma_1}])$ and $\text{Spec}(\mathbb{C}[S_{\sigma_2}])$.

- The gluing maps are the identity maps.

We need to check that the above data is sufficient to define a scheme for gluing.

Proposition 4.1.2

Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Then X_Σ is a normal, separated toric variety.

Proof

Sketch proof:

Since each cone in Σ is strongly convex, we know that $\{0\} \in \Sigma$ and it is the face of every cone. Hence we have inclusions $\text{Spec}(\mathbb{C}[S_{\{0\}}]) \hookrightarrow \text{Spec}(\mathbb{C}[S_\sigma])$ for every $\sigma \in \Sigma$. Hence each of the torii is glued by the identity. Thus $\text{Spec}(\mathbb{C}[S_{\{0\}}])$ is a torus contained in X_Σ . For each $\sigma \in \Sigma$, we know that $\text{Spec}(\mathbb{C}[S_{\{0\}}])$ acts on $\text{Spec}(\mathbb{C}[S_\sigma])$ for each $\sigma \in \Sigma$. Hence they glue to an action on X_Σ . (Check: the action is compatible with intersections and patches).

Since each $\text{Spec}(\mathbb{C}[S_\sigma])$ is irreducible, we must have X_Σ is irreducible. Hence X_Σ is a toric variety.

Since normality is a local condition, we conclude that X_Σ is normal.

Now notice that X_Σ is separated if and only if for any cones $\sigma_1, \sigma_2 \in \Sigma$, the product map $\Delta : \text{Spec}(\mathbb{C}[S_{\sigma_1 \cap \sigma_2}]) \rightarrow \text{Spec}(\mathbb{C}[S_{\sigma_1}]) \times \text{Spec}(\mathbb{C}[S_{\sigma_2}])$ is Zariski closed. Now the map Δ is induced by the $\mathbb{C}$ -algebra homomorphism $\Delta^* : \mathbb{C}[S_{\sigma_1}] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma_2}] \rightarrow \mathbb{C}[S_{\sigma_1 \cap \sigma_2}]$ by $\chi^{m_1} \otimes \chi^{m_2} \mapsto \chi^{m_1 + m_2}$. By the separation lemma, we know that $S_{\sigma_1} + S_{\sigma_2} = S_{\sigma_1 \cap \sigma_2}$. Thus Δ^* is surjective. Then we have

$$\mathbb{C}[\text{im}(\Delta)] = \mathbb{C}[S_{\sigma_1}] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma_2}] / \ker(\Delta^*)$$

so that $\text{im}(\Delta)$ is Zariski closed. ■

Theorem 4.1.3 ((Sumihiro))

Let X be a normal separated variety equipped with the action of a torus T . Let $p \in X$. Then there exists an affine open neighbourhood U of p such that U is T -invariant.

Corollary 4.1.4

Let X be a normal separated toric variety. Then there exists a fan Σ in $\mathbb{R}^n$ for some $n \in \mathbb{N}^+$ such that $X = X_\Sigma$.

Intuition: Adding a new cone to a fan partially compactifies.

Proposition 4.1.5

Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Then the following are true.

- X_Σ is smooth if and only if σ is smooth for all $\sigma \in \Sigma$.

- X_Σ is an orbifold if and only if σ is simplicial for all $\sigma \in \Sigma$.
- Σ is complete if and only if X_Σ .

Example 4.1.6

Let $e_1, e_2 \subseteq \mathbb{R}^2$ be the standard basis vectors. Let Σ be the fan whose maximal cones are given by $\text{Cone}(e_1, e_2), \text{Cone}(e_1, -e_1 - e_2), \text{Cone}(e_2, -e_1 - e_2)$. Then $X_\Sigma = \mathbb{P}^2$.

Proposition 4.1.7

Let $n_1, n_2 \in \mathbb{N}^+$. Let Σ_1, Σ_2 be fans in $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}$ respectively. Then we have

$$X_{\Sigma_1 \times \Sigma_2} \cong X_{\Sigma_1} \times X_{\Sigma_2}$$

4.2 Orbit Cone Correspondence**Definition 4.2.1 (Distinguished Point of a Cone)**

Let $\sigma \subseteq \mathbb{R}^n$ be a cone. Let $\gamma : S_\sigma \rightarrow \mathbb{C}$ be the monoid homomorphism defined by

$$\gamma(m) = \begin{cases} 1 & \text{if } m \in S_\sigma \cap \sigma^\perp = \sigma^\perp \cap (\mathbb{R}^n)^* \\ 0 & \text{otherwise} \end{cases}$$

Define the distinguished point of σ to be the closed point in $\text{Spec}(\mathbb{C}[S_\sigma])$ associated to γ . We denote it by p_σ .

Definition 4.2.2 (Torus Orbits)

Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone. Define the orbit of σ to be

$$O(\sigma) = p_\sigma \cdot T_{X_\Sigma}$$

Lemma 4.2.3

Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone. Then we have

$$\dim(O(\sigma)) = n - \dim(\sigma)$$

Theorem 4.2.4 (The Orbit-Cone Correspondence)

Let Σ be a fan in $\mathbb{R}^n$. Then there is a one to one correspondence

$$\Sigma \xleftrightarrow{1:1} T_{X_\Sigma} \text{ orbits in } X_\Sigma$$

given by $\sigma \mapsto O(\sigma)$.

Proposition 4.2.5

Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone. Then we have

$$\text{Spec}(\mathbb{C}[S_\sigma]) = \bigcup_{\tau \text{ is a face of } \sigma} O(\tau)$$

4.3 Divisors on Toric Varieties**Proposition 4.3.1**

Let Σ be a fan in $\mathbb{R}^n$. Let χ be a character of T_{X_Σ} . Then the following are true.

- χ is a rational function on X_Σ .
- The divisor of χ is given by

$$\text{div}(\chi) = \sum_{\rho \in \Sigma(1)} \langle \chi, u_\rho \rangle \overline{O(\rho)}$$

where u_ρ is a generator for $\rho \cap \mathbb{Z}^n$.

Definition 4.3.2 (*T-Invariant Divisor Group*)

Let Σ be a fan in $\mathbb{R}^n$. Define the T_{X_Σ} -invariant divisor group of X_Σ to be the subgroup

$$\text{Div}_{T_{X_\Sigma}}(X_\Sigma) = \bigoplus_{\rho \in \Sigma(1)} \overline{\mathbb{Z}O(\rho)} \leq \text{Div}(X_\Sigma)$$

Proposition 4.3.3

Let Σ be a fan in $\mathbb{R}^n$. Then the following are true.

- There is an exact sequence of the form

$$\chi(T) \longrightarrow \text{Div}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{Cl}(X_\Sigma) \longrightarrow 0$$

where the first map is given by $\chi \mapsto \text{div}(\chi)$ and the second map is the restriction of the quotient map from the group of divisors to the class group.

- Let u_ρ be a generator for $\rho \cap \mathbb{Z}^n$. If $\{u_\rho \mid \rho \in \Sigma(1)\}$ spans $\mathbb{Z}^n$ (i.e. X_Σ has no torus factors), then there the above sequence is exact on the left:

$$0 \longrightarrow \chi(T) \longrightarrow \text{Div}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{Cl}(X_\Sigma) \longrightarrow 0$$

Example 4.3.4

We have

$$\text{Cl}\left(\text{Spec}\left(\frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}\right)\right) = \mathbb{Z}$$

Proof

Recall that the affine toric variety is the toric variety associated to the cone $\text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3)$. It has four 1-dimensional faces given by $e_1, e_2, e_1 + e_3, e_2 + e_3$. Now the character lattice for the torus is $\mathbb{Z}^3$ with generators e_1, e_2, e_3 . Then we compute that $\text{div}(e_1) = \overline{O(e_1)} + \overline{O(e_1 + e_3)}$ and

etc. By the above proposition, it suffices to compute the cokernel of the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. So

we are done. ■

Example 4.3.5

We have

$$\text{Cl}\left(\text{Spec}\left(\frac{\mathbb{C}[x_0, \dots, x_d]}{(x_i x_{j+1} - x_j x_{i+1} \mid 0 \leq i < j \leq d-1)}\right)\right) = \mathbb{Z}/d\mathbb{Z}$$

Proof

Recall that the rational normal cone is given by the cone $\sigma = \text{Cone}(e_2, de_1 - e_2)$. It has two 1-dimensional faces whose generators are given by e_2 and $de_1 - e_2$. Now the character lattice of S_σ is $\mathbb{Z}^2$ with generators e_1 and e_2 . Then we compute that

$$\text{div}(e_1) = \langle e_1, e_2 \rangle \overline{O(e_2)} + \langle e_1, de_1 - e_2 \rangle \overline{O(de_1 - e_2)} = d\overline{O(de_1 - e_2)}$$

and similarly that $\text{div}(e_2) = \overline{O(e_2)} - \overline{O(de_1 - e_2)}$. By the above proposition, it suffices to compute the cokernel of $\begin{pmatrix} 0 & 1 \\ d & -1 \end{pmatrix}$. Hence we are done. ■

Definition 4.3.6 (*T-Invariant Cartier Divisor Group*)

Proposition 4.3.7

Let Σ be a fan in $\mathbb{R}^n$. Then the following are true.

- There is an exact sequence of the form

$$\chi(T) \longrightarrow \text{CDiv}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{CCl}(X_\Sigma) \longrightarrow 0$$

where the first map is given by $\chi \mapsto \text{div}(\chi)$ and the second map is the restriction of the quotient map from the group of divisors to the class group.

- Let u_ρ be a generator for $\rho \cap \mathbb{Z}^n$. If $\{u_\rho \mid \rho \in \Sigma(1)\}$ spans $\mathbb{Z}^n$ (i.e. X_Σ has no torus factors), then there the above sequence is exact on the left:

$$0 \longrightarrow \chi(T) \longrightarrow \text{Div}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{CCl}(X_\Sigma) \longrightarrow 0$$

Proposition 4.3.8

Let $\sigma \subseteq \mathbb{R}^n$ be a strongly convex rational polyhedral cone. Then the following are true.

- Every T_{X_Σ} -invariant Cartier divisor on $\text{Spec}(\mathbb{C}[S_\sigma])$ is the divisor of a character.
- $\text{Pic}(\text{Spec}(\mathbb{C}[S_\sigma])) = 0$.

Proposition 4.3.9

Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone such that $\dim(\sigma) = n$, then $\text{Pic}(X_\Sigma)$ is a free abelian group.

Proposition 4.3.10

Let Σ be a fan in $\mathbb{R}^n$. Then the following are equivalent.

- X_Σ is smooth.
- Every Weil divisor is a Cartier divisor.
- $\text{Cl}(X_\Sigma) \cong \text{CCl}(X_\Sigma)$.

Proposition 4.3.11

Let Σ be a fan in $\mathbb{R}^n$. Then the following are equivalent.

- Σ is simplicial.
- X_Σ is an orbifold.
- $\text{CCl}(X_\Sigma)$ has finite index as a subgroup of $\text{Cl}(X_\Sigma)$.

Proposition 4.3.12

4.2.8 Cox

Proposition 4.3.13

4.3.2 Cox

Lmm5.1.1